

# Saying “I’m Not Sure” to Young Learners: A Protocol for Designing and Expert-Reviewing Uncertainty Cues in a Children’s AI Tutor

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*Study protocol. No results yet. Not peer reviewed.*

## Abstract

If an AI tutor should sometimes decline to answer, it also needs a way to say so that a child can understand. Research with adults shows that first-person expressions of uncertainty can reduce overreliance on a language model, and research with children shows that they form views about conversational agents’ intelligence and trust, but how a tutor should express uncertainty to young learners remains largely undesigned. This protocol describes a design study for learners of about ages 8 to 11. I will build a scripted, clickable prototype of one short science lesson in five variants that differ only in the uncertainty cue: no cue, a plain-language hedge, a visual confidence meter, a character who shows it is thinking, and an explicit “I don’t know” with a hand-off to a teacher. The variants will be evaluated without children, through a heuristic review and walkthroughs by three to five adult experts. The output is a set of candidate cues and design risks, ready for a later study with children under ethics review.

**Keywords:** uncertainty communication; children; AI tutors; abstention; trust calibration; heuristic evaluation; prototyping

## 1. Introduction

My earlier work found that a retrieval-augmented question-answering system never declined to answer, even on questions it had no basis for (Choi, 2026). If a tutor *should* sometimes say it does not know, a second question follows: how should it say it, especially to a child?

Children are a demanding audience for this. A cue must be understandable at their reading level, it should invite a useful next step such as checking a book or asking a teacher, and it should not make the tutor seem broken or untrustworthy in general. This study designs candidate cues and gathers expert critique before any child is involved.

*How can an AI tutor tell a young learner that it is unsure in a way that prompts checking rather than blind acceptance or blanket distrust?*

## 2. Background

Druga et al. (2017) studied how 26 children aged 3 to 10 interacted with four agents, including Amazon Alexa and Google Home, and asked them about trust, intelligence, and personality; they identified four themes, including perceived intelligence and understanding. Ruan et al. (2020) found, with 72 children in grades 3 to 5, that embedding a math task in a narrative boosted engagement and that adding a tutoring chatbot improved learning outcomes. With adults, Kim et al. (2024) found in a pre-registered experiment (N = 404) that first-person uncertainty expressions such as “I’m not sure, but...” decreased agreement with a fictional LLM-infused search engine and increased participants’ accuracy. Heuristic evaluation (Nielsen & Molich, 1990) offers a way to find interface problems without end users, which suits an early design stage.

**Table 1.** Prior work and what this study adds.

| Work                    | Finding                                                           | What it leaves open                                |
|-------------------------|-------------------------------------------------------------------|----------------------------------------------------|
| Druga et al. (2017)     | Children form views of agents' intelligence and trust             | How agents should signal their limits              |
| Ruan et al. (2020)      | A narrative tutor with chatbot help supported children's learning | The tutor's own uncertainty as a design variable   |
| Kim et al. (2024)       | First-person uncertainty reduced adults' overreliance             | Whether similar cues work for children             |
| Nielsen & Molich (1990) | Heuristic evaluation finds problems without end users             | Heuristics specific to children and AI uncertainty |

### 3. Research questions

1. **RQ1.** What forms can an AI tutor's uncertainty take for young learners, in words, visuals, character behavior, or a hand-off to a person?
2. **RQ2.** Which forms do experts judge most likely to be understood by children and to prompt checking rather than blind acceptance?
3. **RQ3.** What risks do experts see, such as children losing trust entirely or concluding that the tutor is broken?

### 4. Method

#### 4.1 Design space

Each variant answers the same questions in the same lesson, so the cue is the only thing that changes (Table 2).

**Table 2.** Uncertainty cue variants.

| Variant             | Cue                            | What the child sees or hears                                   |
|---------------------|--------------------------------|----------------------------------------------------------------|
| A. No cue (control) | None                           | A confident answer                                             |
| B. Plain words      | First-person hedge             | "I'm not sure about this one, but I think..."                  |
| C. Visual meter     | Confidence picture             | A small gauge icon next to the answer                          |
| D. Character        | The guide shows it is thinking | The guide character pauses and suggests checking the book      |
| E. Hand-off         | Abstain and redirect           | "I don't know. Let's ask your teacher or look it up together." |

#### 4.2 Prototype

The prototype is scripted rather than driven by a live model, so every variant shows identical answers. It covers one short lesson (for example, why leaves change color) with six questions: three the tutor answers correctly, two it answers incorrectly, and one it cannot answer. It will be built as a clickable Figma prototype or a single web page with a variant switcher. The guide character will be an original drawing made for this study.

#### 4.3 Evaluation

1. **Heuristic review.** I will score each variant against Nielsen's heuristics plus three written for this problem: the cue is understandable at the target reading level; the cue invites a next action, such as checking; and the cue does not make the tutor seem broken.

2. **Expert walkthroughs.** Three to five adult experts (teachers or HCI researchers) will each click through all variants in a randomized order for about 30 minutes and answer the same questions for each: would a child understand this, what would the child do next, and what could go wrong.

**Open item.** Expert walkthroughs may count as human subjects research. Before inviting experts, I will ask my university’s IRB office whether a determination is needed. If the answer comes too late, I will run the heuristic review alone and report it as a single-author review.

## 5. Analysis plan

- **RQ1:** the design space itself, documented with screenshots of each variant.
- **RQ2:** heuristic scores per variant, and expert judgments summarized by variant with counts.
- **RQ3:** expert comments coded by theme; every reported theme names how many experts raised it.

## 6. Ethics and limitations

No children take part in this study. Any later study with children will require ethics review and parental consent.

- Adult judgments about children are predictions, not evidence of how children respond.
- A scripted prototype cannot show how cues behave with a real model’s errors.
- Three to five experts give direction, not generalizable findings.

## 7. Timeline

**Table 3.** Planned schedule (2026).

| Stage                                    | Dates            |
|------------------------------------------|------------------|
| Ask the IRB office and invite experts    | Oct 12 to Oct 18 |
| Script and design the variants           | Oct 12 to Oct 25 |
| Build the prototype                      | Oct 26 to Nov 8  |
| Heuristic review and expert walkthroughs | Nov 9 to Nov 22  |
| Writing the paper                        | Nov 23 to Nov 29 |
| Posting the paper and prototype          | Nov 30           |

## 8. Declarations

**Origin of this work.** This study extends the abstention problem described in Choi (2026) to the design of uncertainty cues for children.

**Use of AI tools.** I used Claude (Anthropic), an AI assistant, to verify references against Crossref records and to draft text and tables.

**Funding.** None.

**Competing interests.** None.

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